HIROSHIMA UNIVERSITY · SIP PHASE 3 SMART MOBILITY PLATFORM

Activity-based Travel Demand Modeling with a Dynamic Discrete Choice Model

How people's **daily activities and travel** respond to the city and to policy

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Machigurumi Simulator progress seminar · 23 October 2026

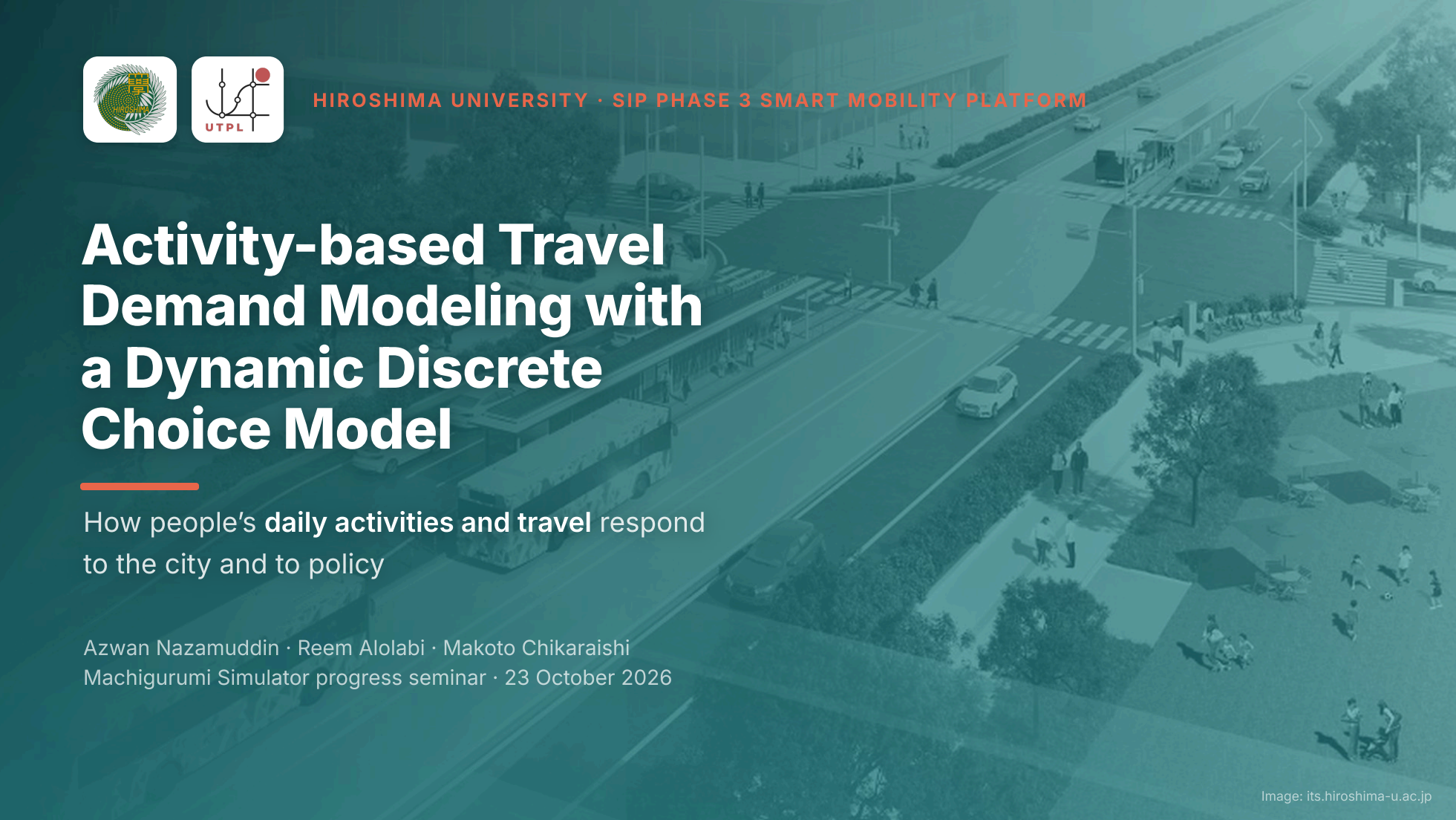
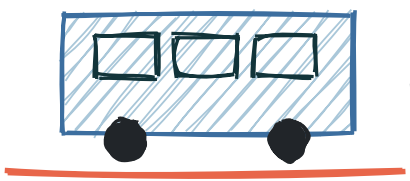


Image: its.hiroshima-u.ac.jp

If a BRT opens, how will behaviour change?

Before a new BRT line is built, planners need to know how people will respond: more than a change of mode.



**a new BRT line,
or another new mode**

MODE	**Who switches to the bus, and from which mode?**
TIMING	**Do people leave home or work at different times?**
DESTINATION	**Do they shop or spend free time in different places?**
ACTIVITIES	**Do they add a stop, combine trips, or drop one?**
ACCESSIBILITY	**What can each person now reach, and who is left out?**

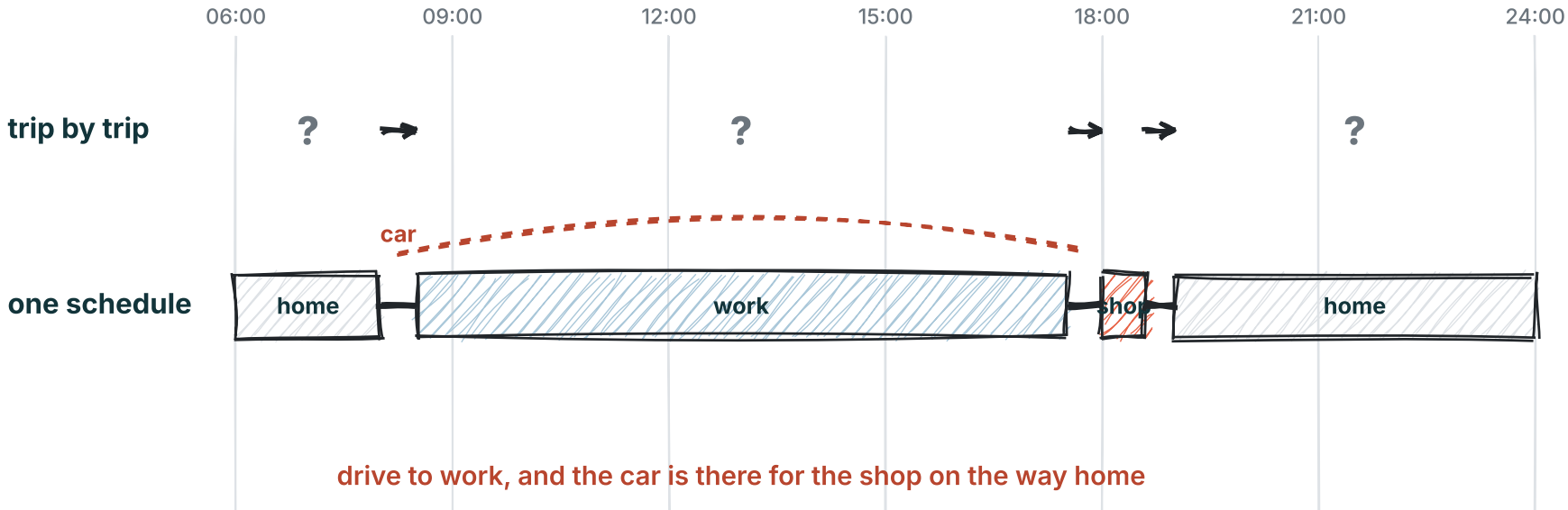
The answers are connected, and they differ from person to person:
a model has to predict them together, for each person.

PART 1

Why activity-based models

Trips are linked

A **trip-based model** predicts each trip alone; an **activity-based model** starts from the day's activities, so one choice changes the next.



PART 2

From traditional models to a dynamic discrete choice model

How models build the same schedule

Four demand models build the same day, one at a time. Last: MATSim, where the day plans go.



5 · MATSim: where plans go

Not a demand model: it takes each day plan into traffic and refines times and routes.
The plan itself comes from a demand model.

How a choice model works

A **discrete choice model**: each option gets a score from its time and cost; better scores are more likely, never certain.



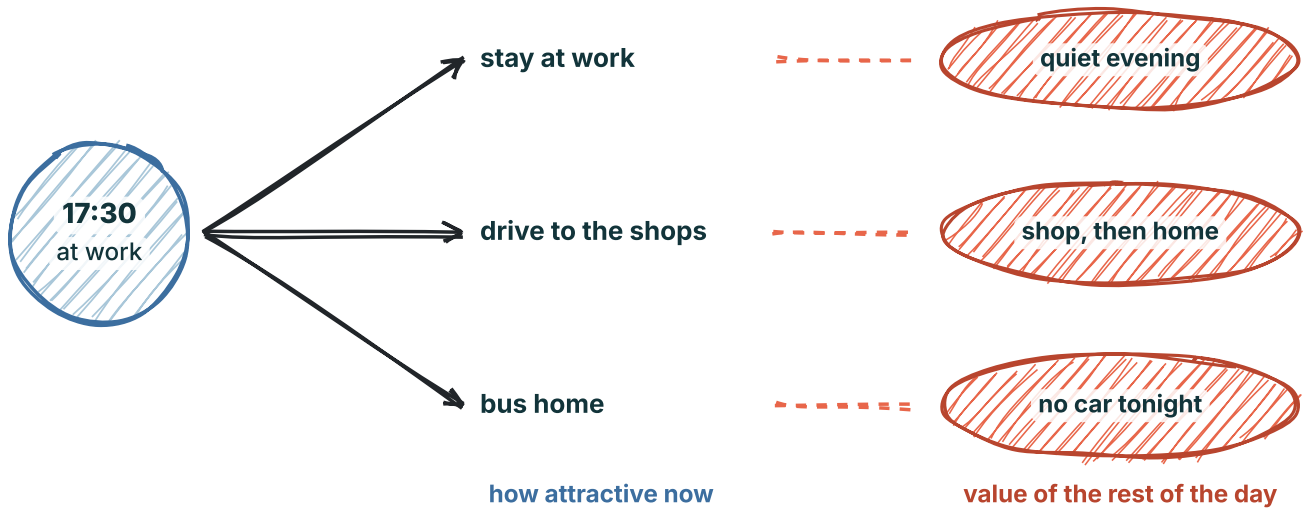
the more attractive an option, the more likely, but never certain

how much people dislike time and cost (their preferences) is learned from diaries

illustrative numbers

From one choice to a whole day of choices

A **DDCM** applies the same idea at every moment: a choice is worth what it gives now, plus what it leaves for later.



The catch: it is heavy to compute

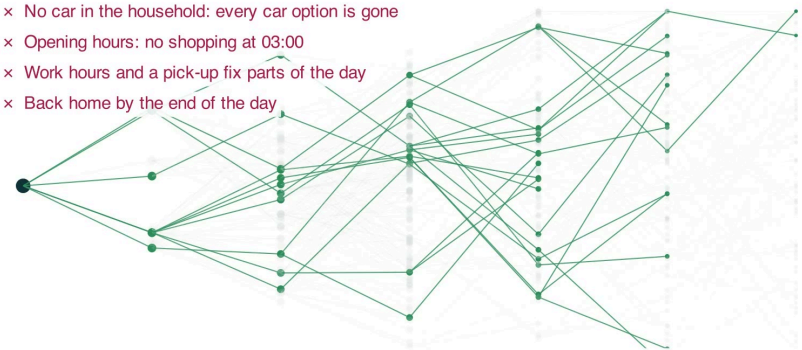
To value one choice, the model values every situation after it, and estimation repeats this hundreds of times.

▶ Animation in the live talk: final frame shown

But one person cannot be in most of them

× Travel takes time: far zones cannot be reached in 15 minutes
× No car in the household: every car option is gone
× Opening hours: no shopping at 03:00
× Work hours and a pick-up fix parts of the day
× Back home by the end of the day

situations that remain
≈ 300,000



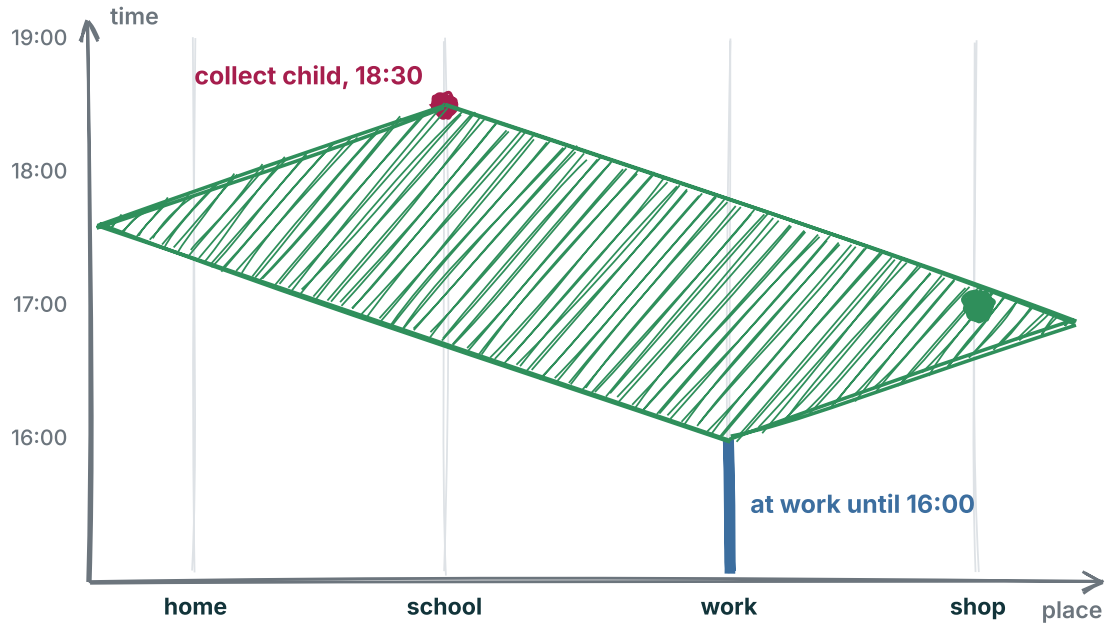
Constraints do two jobs at once:
they describe what a person can do, and they decide how much there is to compute.
But they differ from person to person, which we come back to shortly.

PART 3

People face different constraints

What a person can do: the space-time prism

Constraints limit what a person can do. The **space-time prism** (Hägerstrand) draws every place they can reach.



by car: the prism is wide

the shop is inside it

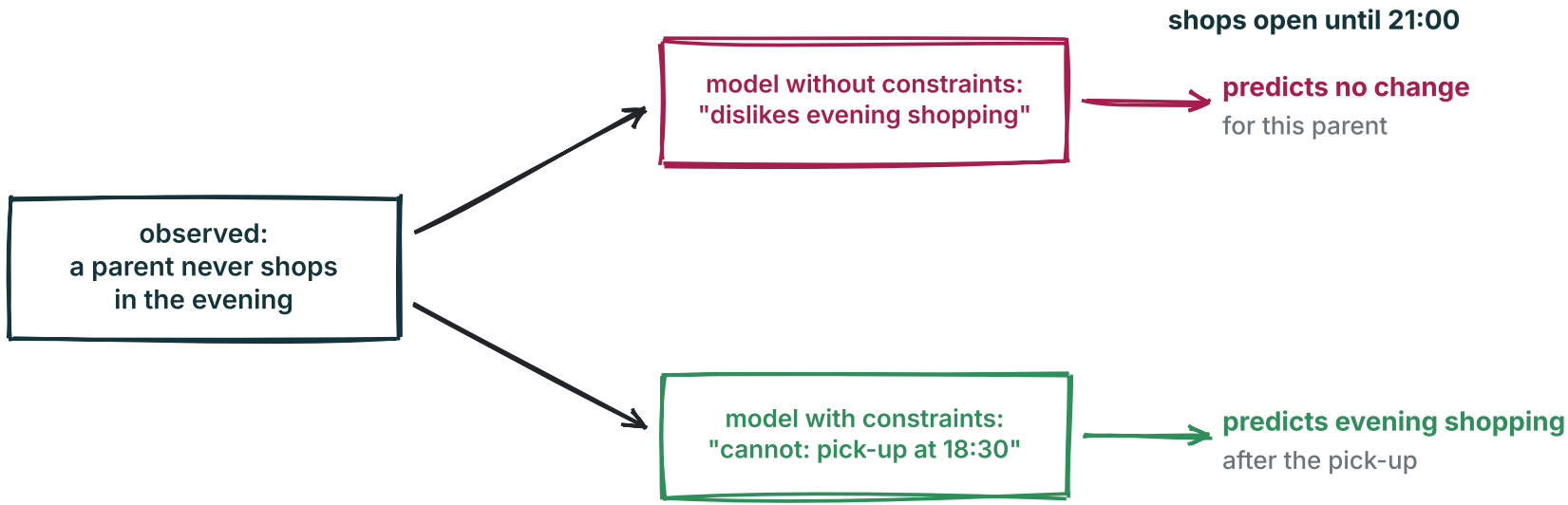
same city, same shop

the prism = every place this person can be, given fixed times and travel speed

(Hägerstrand, 1970)

Ignore constraints, and they read as preference

A diary shows what people did, not why. Without constraints, "cannot" is read as "does not like".



the wrong reading spreads into predicted behaviour change and into accessibility

Why constraints matter for policy: welfare

Accessibility (logsum, in yen) is the value of all the days a person can still live: only as right as their constraints.

Policy question	Constraints ignored	Each person's constraints kept
Shops open until 21:00: does the parent gain?	No gain: "dislikes evening shopping"	**Gains: can shop after the 18:30 pick-up**
A faster bus: how much is it worth to a person without a car?	Valued over days this person cannot live	**Valued only over days this person can live**
Who gains, who is left out?	Differences read as taste; winners blur	**Gains by car, job and home zone (Part 6)**

Misread constraints give the wrong winners and losers. That is why they are kept exact, for everyone.

Exact constraints for every person are heavy too

Each person's own constraints change what every option is worth, so each person needs a full calculation.

Aiko
car, flexible hours

Kenta
no car, works 9-17, child at 18:30

Hana
retired, no car, bus only

three people, three prisms

the same shop at 18:00 is

worth a lot
shop, then drive home

its own calculation

worth nothing
he must be at the pick-up

its own calculation

worth less
a long bus ride home after

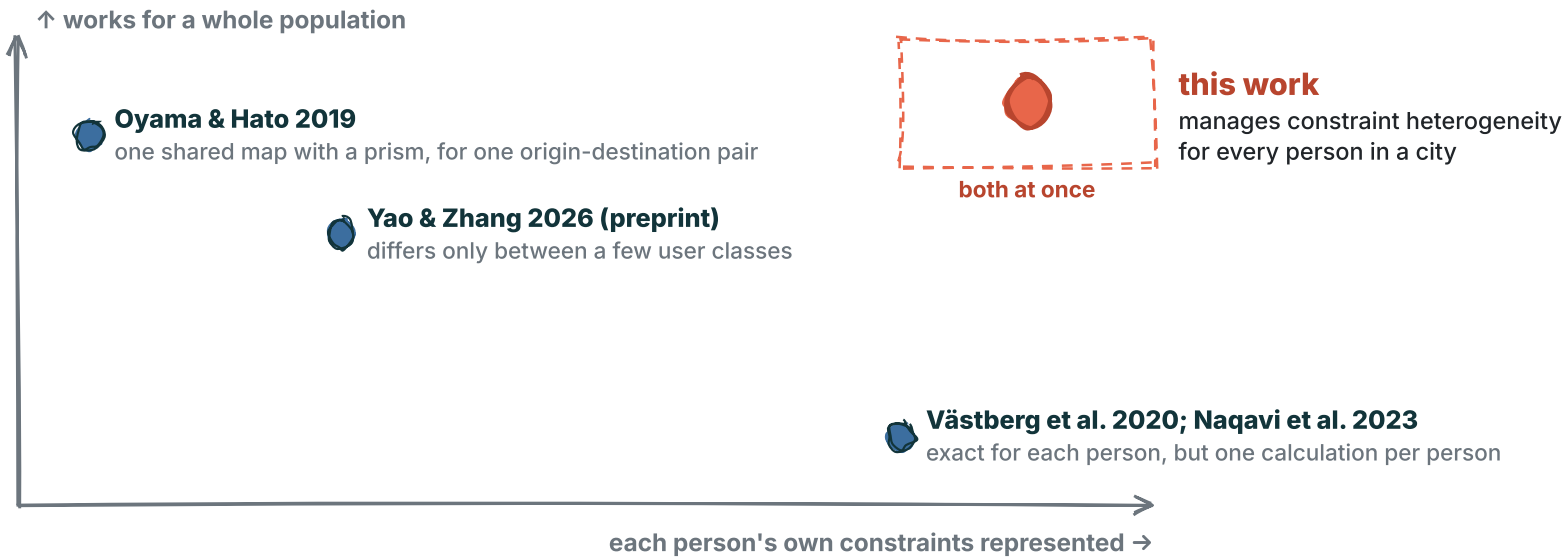
its own calculation

What an option is worth depends on what the person can still do afterwards: nothing can be shared.

× 6,451 people × every try during estimation: constraints end up misread, or too expensive

Where this work sits

Two things at once: each person's own constraints, exactly, for a whole population. Research has each, separately.



Each half is achieved somewhere; managing heterogeneity properly needs both at once.

PART 4

Our framework and approach

Objectives, and where each is shown

Objective	Shown in
1 · **Model each person's whole day**, estimated from local travel diaries, with every person's constraints kept exact	Part 5 Results
2 · **Manage constraint heterogeneity** at city scale: every person keeps their own constraints, at the cost of a handful of shared maps	Part 4 Framework
3 · **Show what a policy does, person by person**: activities, timing, destinations, modes, and who gains accessibility	Part 6 Policy
Link to the simulator: the day plans in MATSim	Part 7

Objective 2 is what makes objective 3 trustworthy: a policy's gain is counted over the days each person can actually live.

Two slots: what people want, what people can do

Each choice has two parts: can this person do it (their own constraints), and how much people like it (shared, from diaries).

$$P(a \mid s, \text{person } n) \propto \delta_n(s, a) \times \exp(v(s, a))$$

feasibility for this person

1 = can, 0 = cannot

workplace and hours, car, opening hours, home by the end

preference

shared, estimated from diaries

a zero from δ means cannot, not does not like:

that is what stops a model reading constraint as preference

Our four constraint types

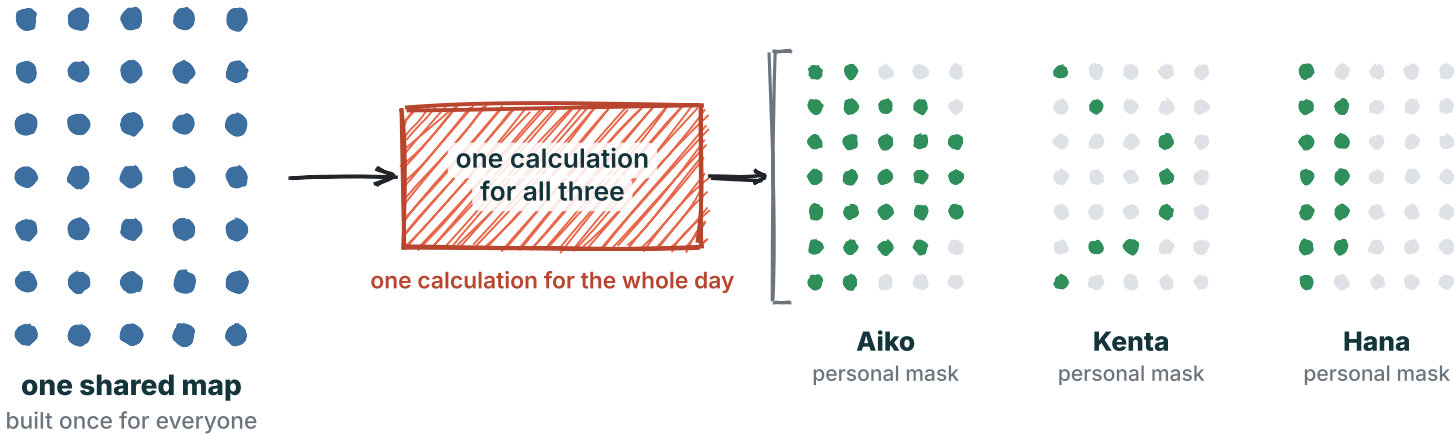
Our classification: four constraint types, sorted by what the rule depends on. The type decides how the model handles it.

type	the rule depends on	for example	what the model does
Institutional **the same for everyone**	*nothing about the person*	opening hours, the network, the day ends at home	→ **removed once, for everyone**
Capability **the means a person has**	*a coarse attribute*	car ownership, which modes are available	→ **one of a few shared maps**
Anchor **where and when one must be**	*a detailed attribute*	home, workplace, work and school hours	→ **a mask on the shared map**
Interdependent **set by another person**	*another person's choice*	the household car, who collects the child	→ **open: needs a household model**

Institutional, capability and anchor constraints are handled exactly on a handful of shared maps; interdependent constraints are a different problem, left open.

One shared map, a mask for each person

One shared map, calculated once; each person's prism is that map with their impossible options switched off. Exact.



Each prism is the shared map with that person's impossible options switched off.

Nothing is approximated: the result is the same as calculating every person separately.

However many people: a few shared maps, and one mask per person.

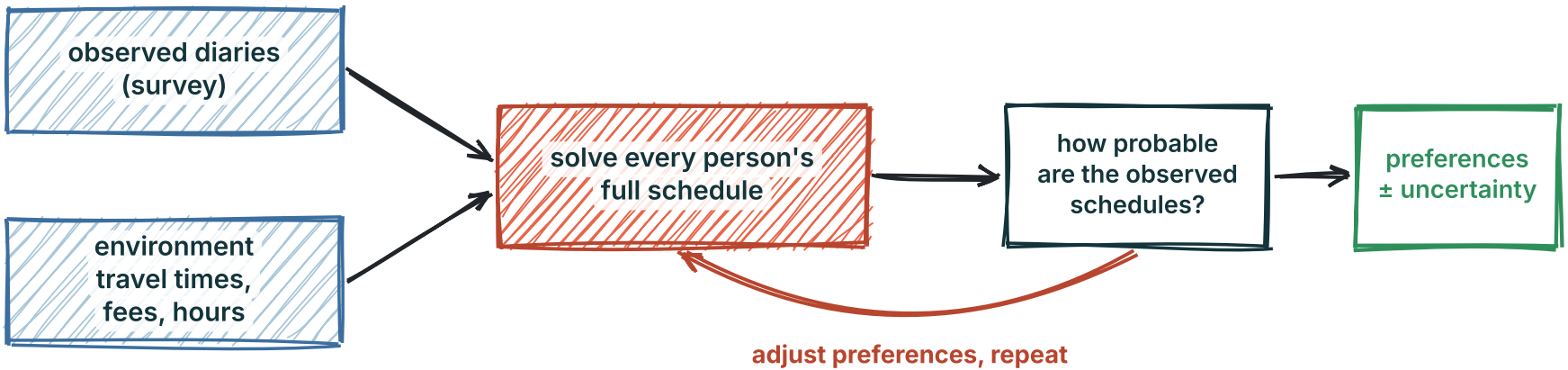
PART 5

Results

Current specification: one iteration, successfully estimated, not the final model

Estimated from real diaries

Estimated from 6,451 people's weekday diaries, with real travel times, parking fees and opening hours.



6,451
persons aged 13+ in the weekday diaries

70
preference parameters, estimated together

144
zones, door-to-door bus and rail times

Estimation results: the parameters

63 of 70 parameters are significant with robust standard errors, and the estimate converged.

Parameter	Estimate	SE	Robust SE	t (robust)
Mode constants				
Car, driver	−5.661	0.033	0.039	−144.6
Car, passenger	−7.083	0.038	0.049	−143.5
Walk	−6.556	0.052	0.065	−101.0
Bicycle	−6.966	0.049	0.052	−134.8
Train	−6.867	0.056	0.084	−82.0
Bus	−8.851	0.099	0.135	−65.4
Travel time (per minute) and cost				
Car time (reference)*	−0.0133		*held fixed: the reference*	
Walk time, relative to car	−0.0024	0.0023	0.0037	−0.7
Bicycle time, relative to car	+0.0102	0.0021	0.0027	3.8
Bus and train time, relative to car	+0.0294		*fixed: 20.4 ¥/min*	
Cost (per yen)	−0.00076	0.00007	0.00010	−7.3

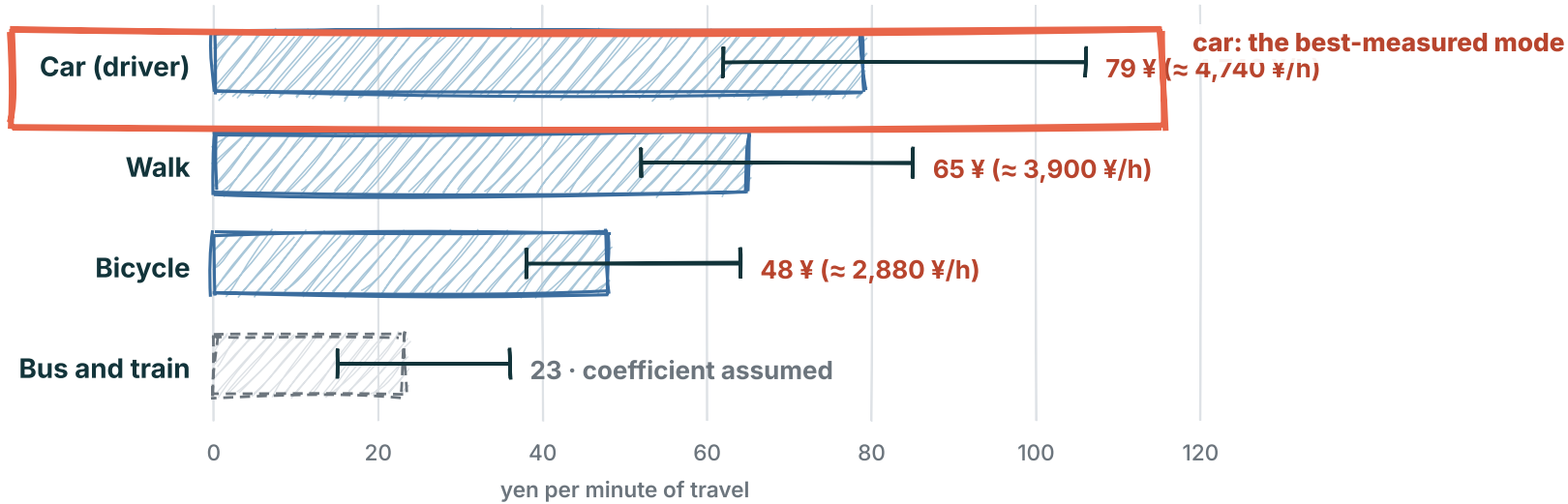
Log-likelihood −40,376.1 · 6,451 persons · 70 of 78 free · exact Hessian · robust (sandwich) SE

Parameter	Estimate	SE	Robust SE	t (robust)
Destinations				
Shopping stop	−2.042	0.085	0.109	−18.8
Leisure stop	−2.796	0.072	0.082	−34.0
Shopping after work	−1.663	0.110	0.118	−14.1
Leisure after work	−1.389	0.096	0.109	−12.7
Size: shops in the zone (log)	0.592	0.014	0.011	55.7
Size: leisure places (log)	0.576	0.013	0.013	44.8
Leisure place open	0.00015	0.00016	0.00017	0.9
Value of a minute, by hour (54 values)				
At home, 18 hours			18 of 18 significant	
Shopping, 18 hours			16 of 18 significant	
Leisure, 18 hours			15 of 18 significant	

***Car time is the reference.** One time value must be held fixed; car time is held where the free estimate stopped. Car's 79 ¥/min is still estimated.

People value travel time differently by mode

A minute in the car is worth about 79 yen; walking 65, cycling 48. Bus and train, 23, is assumed.



Bar: estimate; line: 90% range (robust). Bus and train: assumed from a trip-level model (20.4 there; 23 against time at home), next slide.

One assumption: the public transport time value

Bus and train time is fixed at 20.4 ¥/min from a trip-level model: in the national range, on the low side.

Why it is fixed

- Reported bus and train times are noisy: **±19 min** on the same zone pair
- Time and the train constant cannot be told apart (correlation −0.89)
- Left free, it drifts to **6 ¥/min**

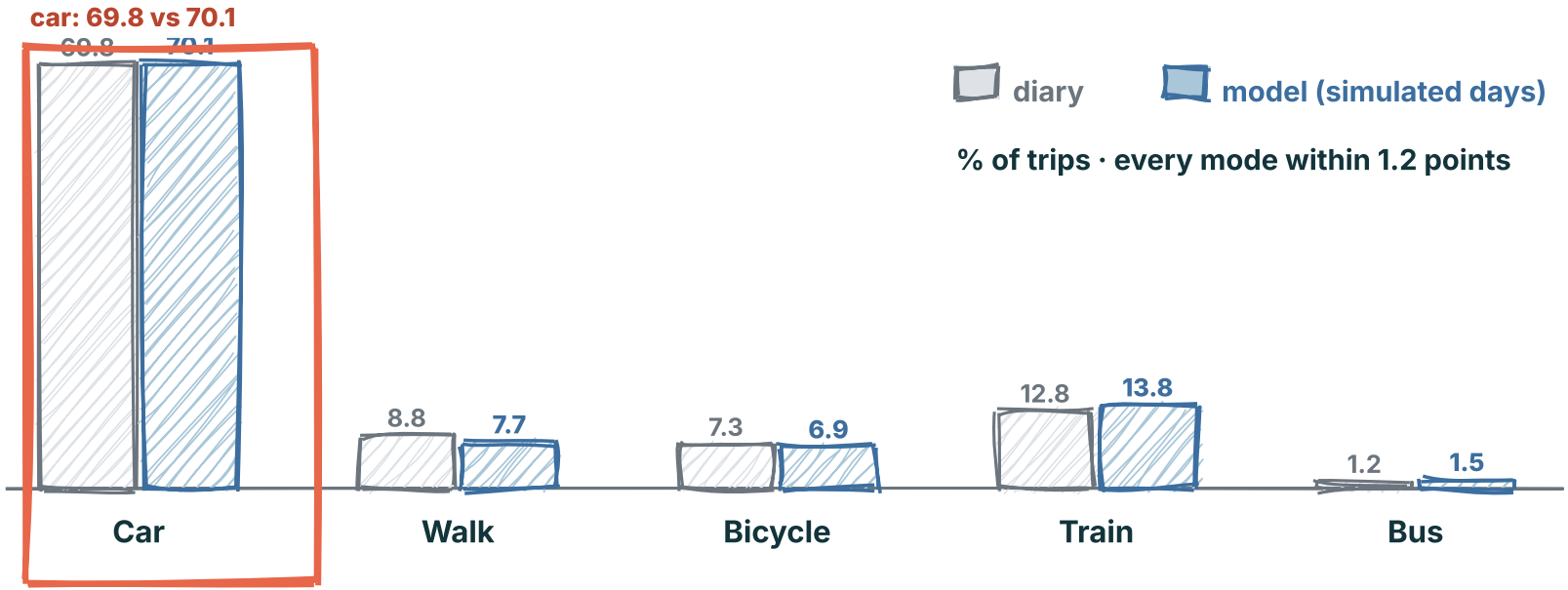
Is 20.4 reasonable?

- **Same diaries and travel times** (15,395 trips)
- **Inside the national range**, 15–38 ¥/min
- **On the low side**: BRT gain, if anything, understated
- **At 13 or 28 ¥/min**: BRT gain 0.7–1.5 ¥ per person-day

Against time at home, 20.4 becomes 23 yen per minute, the bar on the value-of-time slide.

Mode choice is reproduced

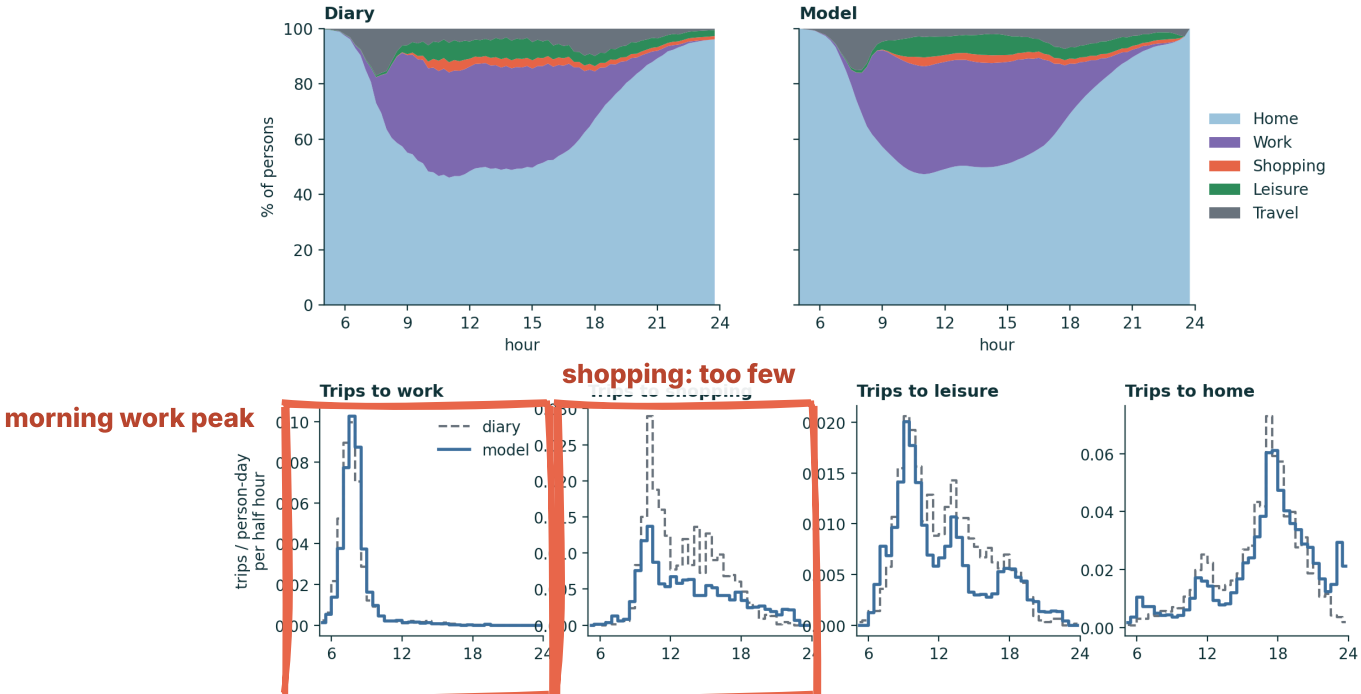
The model fits on mode choice: car about 70% in both, every mode within 1.2 points.



Full sample, 6,477 persons × 3 days. Helped by the fixed reference values; specification still being checked.

The rhythm of the day: fits, except shopping

The shape of the day and the work peak fit; shopping trips are too few, through the whole day.

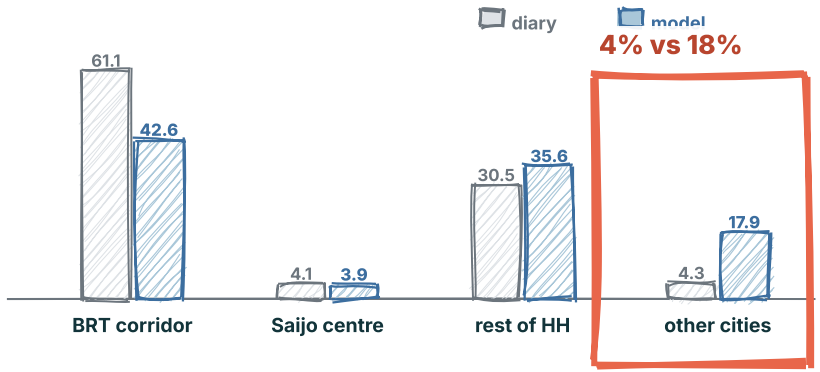


Top: what people are doing every 15 minutes (diary left, model right). Bottom: when trips start (dashed diary, solid model).

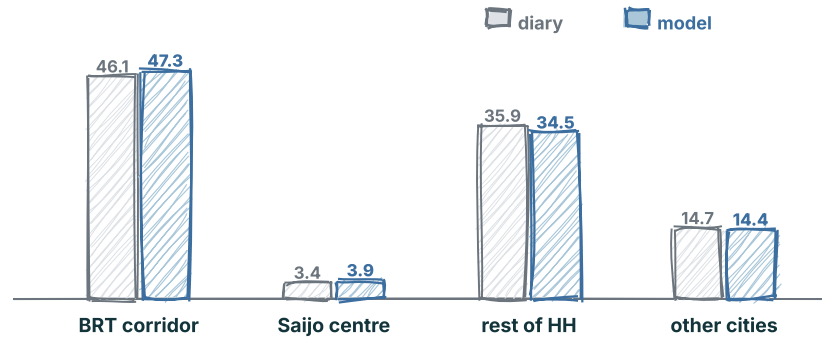
Where people go: leisure fits, shopping does not

Leisure destinations fit within about a point; shopping goes too far: 18% to other cities against 4%.

Shopping, % of trips



Leisure, % of trips

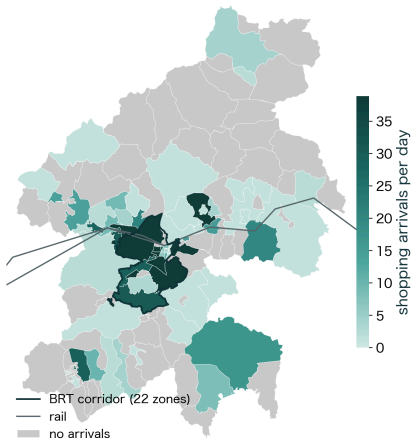


Diary grey, model blue. Corridor: 22 zones within 1.5 km of a stop. Other cities: mostly Hiroshima.

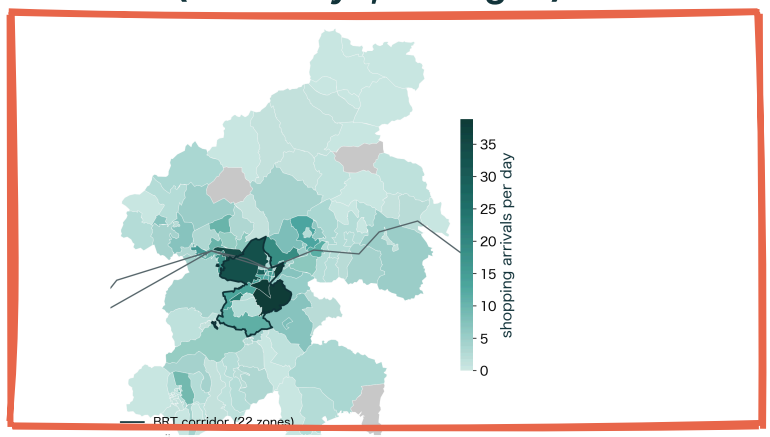
Where shopping trips go: underfitted

Shopping is too spread out: per day, diary shopping reaches 70 of 144 zones, the model about 110.

Diary (one day)



Model (three days, averaged)



spread over almost every zone

Main cause: no store size. A shopping centre counts like one small shop, so trips leak to small zones. Logit keeps every zone some chance.

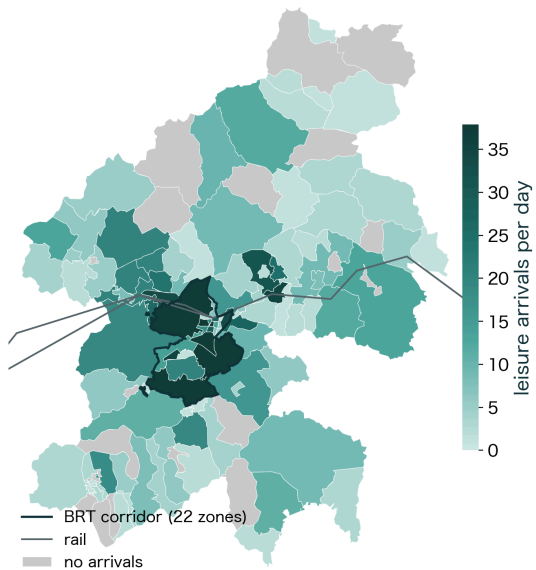
Store data incomplete. 23 zones without opening hours count as always open. Grey diary zones: mostly sampling, not closed shops.

Top 10 zones: 69% of diary shopping, 42% of the model's. Next: store size; later, exact zeros (another entropy).

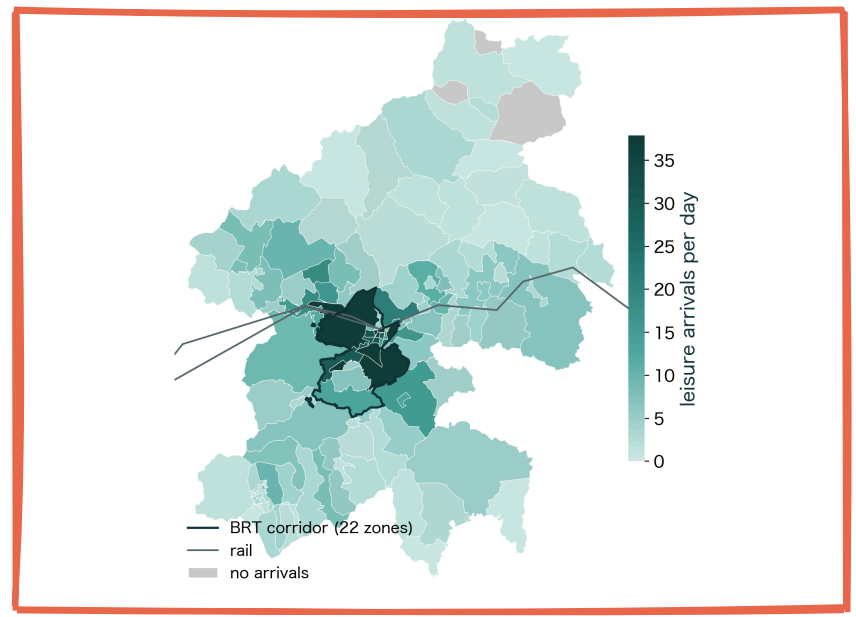
Where leisure trips go: a better fit

Leisure fits better: the diary itself is spread widely, so few zones are empty in either.

Diary



3 empty zones, against 22 in the diary

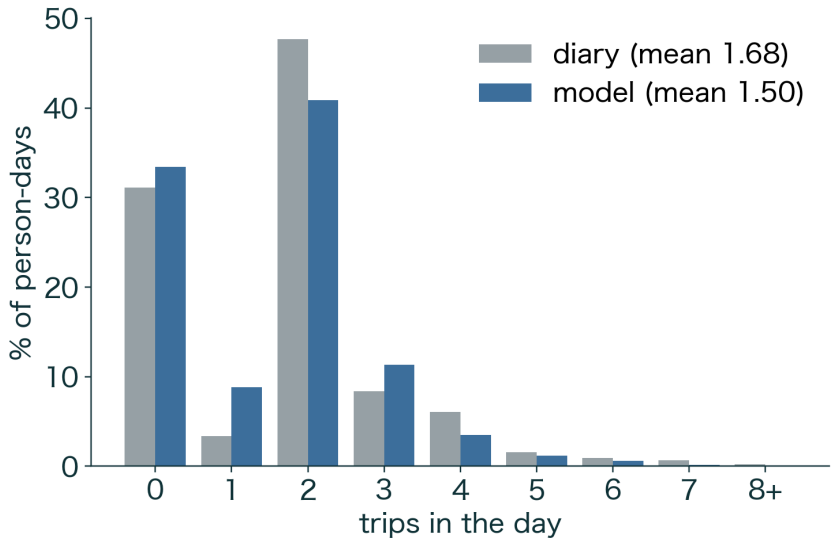


Leisure arrivals per day, same scale. Grey: no arrivals (22 of 122 zones in the diary, 3 in the model).

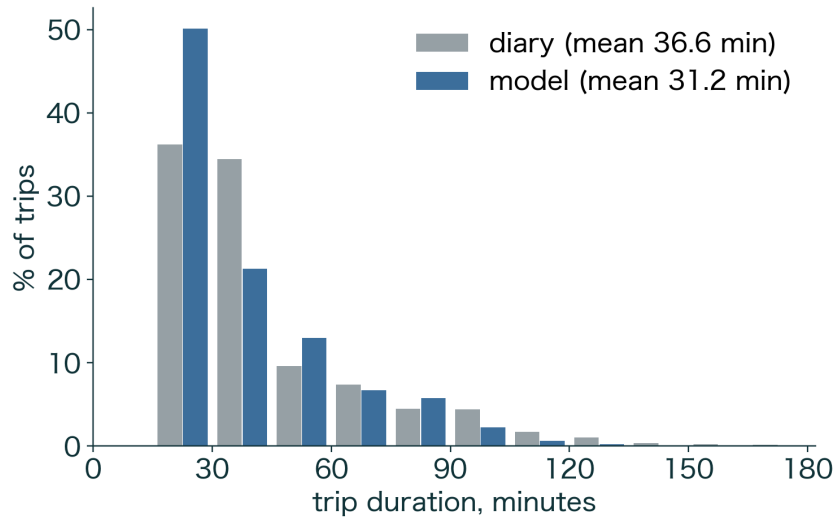
How many trips, and how long: shopping again

Again the weak part is shopping: non-workers shop on 14% of days, against 24% in the diary.

Trips in a day



How long a trip takes



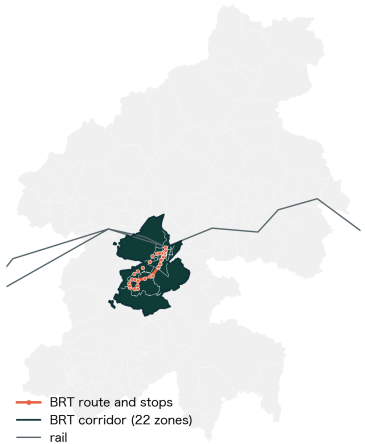
	Days with shopping, non-workers	Days with leisure, non-workers	Length of a shopping visit
Diary	24%	27%	87 min
Model	14%	28%	120 min

PART 6

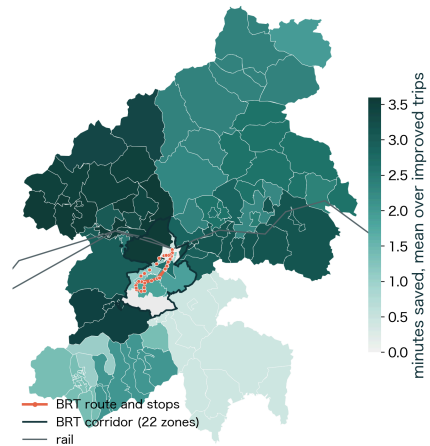
Policy use

What if we add a BRT?

The BRT saves up to about 4 minutes a trip, also off the corridor: it feeds Saijo station.



The corridor: 35 stops; 22 zones within 1.5 km of a stop, home to 2,143 sample residents.



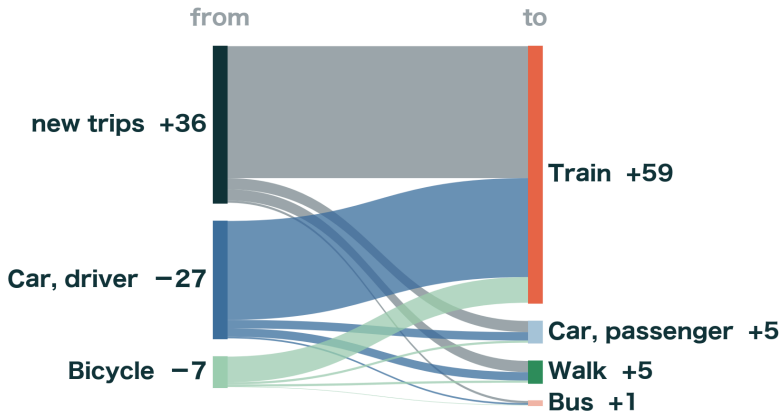
Minutes saved by origin zone, on the trips the BRT makes faster (820 zone pairs).

In the model: faster buses at today's fare (6/15/12 min headways). 480 zone pairs get a faster bus trip, 349 a faster train trip via Saijo.
Not yet: a lane, congestion, comfort, reliability.

BRT: who changes mode

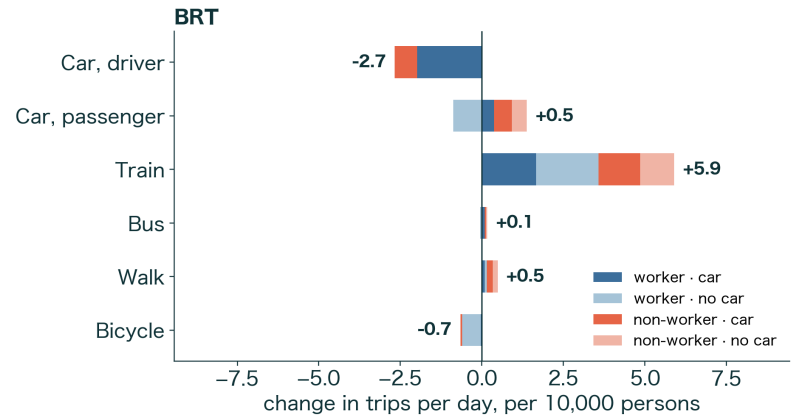
Most new public transport use appears as train trips: the BRT feeds Saijo station.

Where the trips come from, and go to



Trips per day, per 100,000 persons: only the trips that move.

Who changes, by group



Per 10,000 persons, stacked by group.

walk → BRT → walk = a **bus trip**

walk → BRT → Saijo station → train → walk = a **train trip** (main mode)

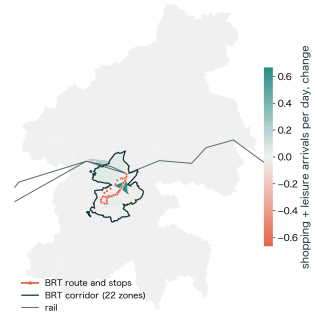
Limits: the BRT is part of the bus network; as a feeder it counts in the main mode, train. So mode shares cannot show BRT use. Mode choice is not yet fully fitted: read the direction, not the size. **Next:** re-specify the modes.

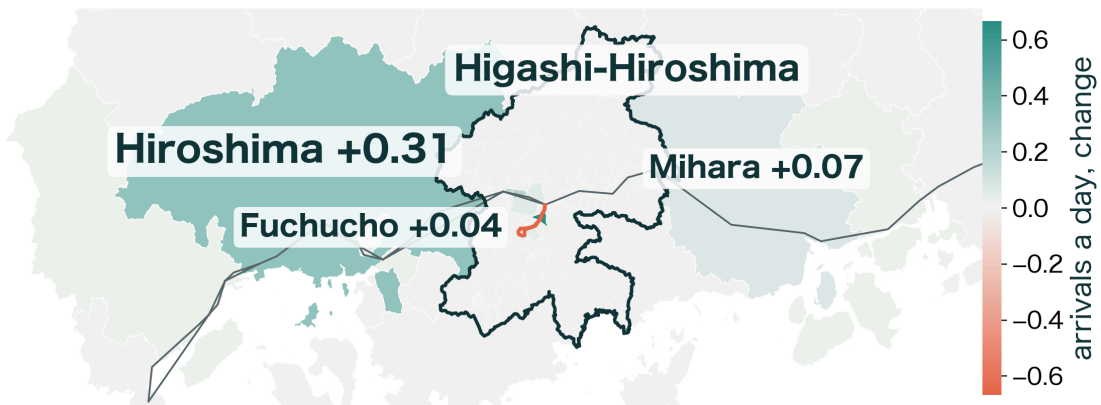
BRT: where free-time trips go

Free-time trips shift a little, mostly by train: to the corridor and to Hiroshima city.

Arrivals a day, change	Corridor	Rest of HH	Hiroshima city	Other cities
Train	**+0.80**	+0.33	**+0.17**	+0.15
Walk	+0.14	+0.02	+0.10	+0.01
Car	+0.07	−0.06	+0.04	−0.00
Share of arrivals today	46%	34%	9%	7%

Bus and bicycle: about 0. Saijo centre (4%) not shown. Below: the wider region; BRT route in coral.



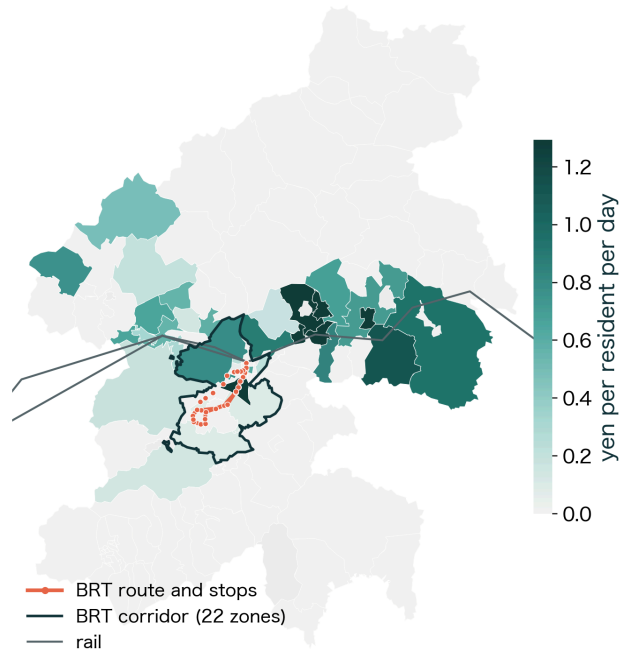


Who gains

People without a car and workers gain most: about 2 ¥ a day, against 0.6 ¥ for car owners.

Yen per person-day	Time saved	Logsum
Workers	**+2.16**	+1.56
No car in the household	**+2.00**	+1.57
Corridor residents	+1.46	+1.16
Elsewhere	+0.86	+0.66
Car owners	+0.56	+0.44
Non-workers	+0.29	+0.32
All persons	**+1.06**	**+0.83**
Share gaining / losing	3.1% / 0%	98.7% / 1.3%

Accessibility, ¥ per person-day: time saved × 20.4 ¥/min; the logsum agrees. Map: by home zone.



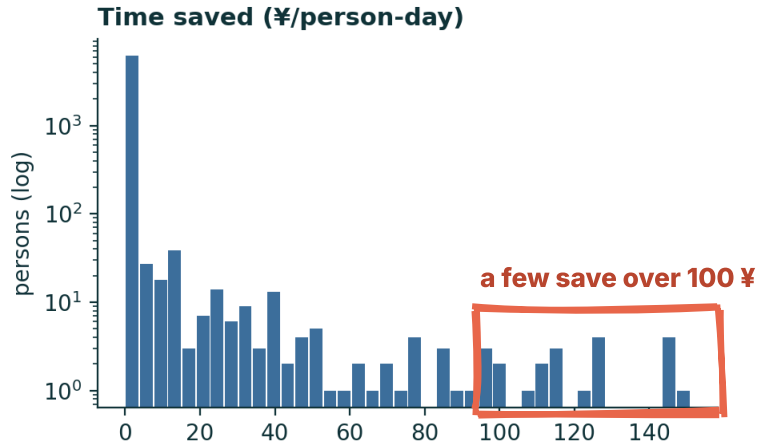
BRT: small on average, large for a few

About 1 ¥ per person a day on average, but 3% of people gain, a few more than 100 ¥.

+1.06 ¥
per person per day, on average

3.1%
of persons gain (about 200 in the sample)

+0.04 pt
bus and train share (15.25% → 15.29%)



Persons (log scale) by yen saved per day: almost everyone at zero, a few save more than 100 yen.

Gain = minutes saved on bus and train × 20.4 ¥/min; full sample, point estimate.

City-wide policies, same people

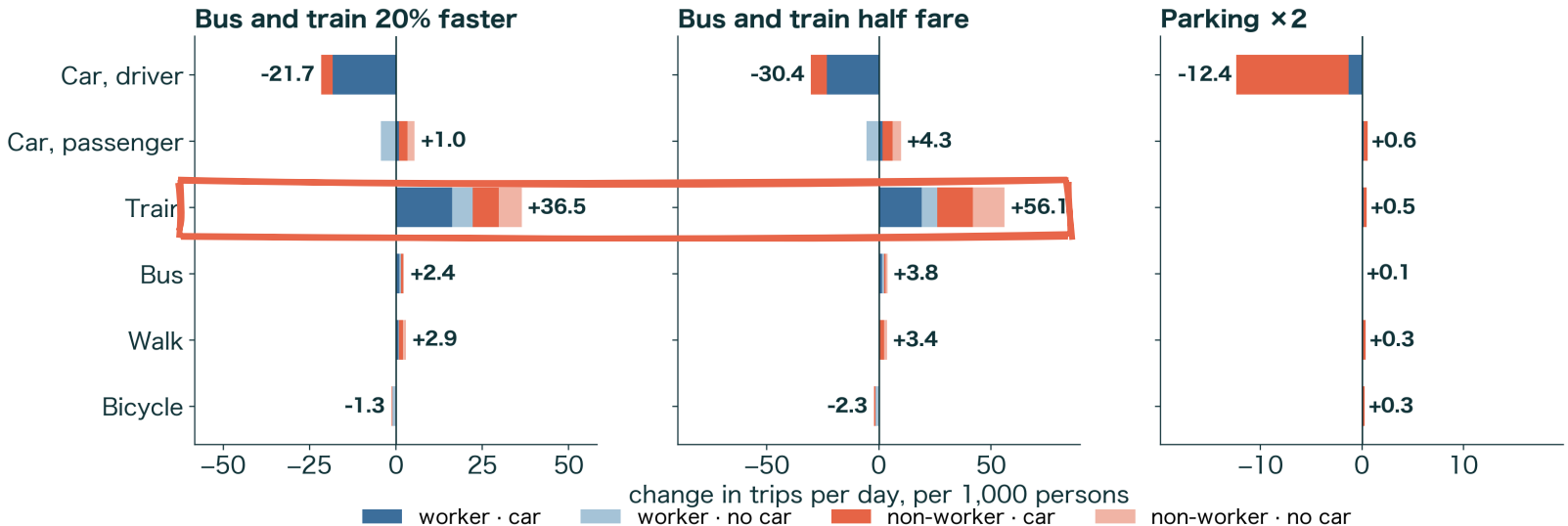
Our BRT only changes corridor travel times. Bigger levers show the model can respond: 60–90 ¥ a day.

Policy	What changes	Accessibility ¥ per person-day	Gain / lose	Car Δ pt	Bus + train Δ pt	Trips per day	Average trip
Faster public transport	every bus and train time ×0.8	**+61**	38% / 0%	−2.22	+2.31	+1.3%	−4.0%
Cheaper public transport	every bus and train fare ×0.5	**+93**	100% / 0%	−3.22	+3.49	+2.3%	+4.2%
Parking +50%	three central zones that charge	−3.9	0% / 66%	−0.17	+0.08	−0.4%	−0.1%
Parking ×2	same three zones	−6.6	0% / 66%	−0.29	+0.14	−0.7%	−0.0%
Work 1 hour later (sensitivity)	every worker's start time	−532	4% / 37%	+0.98	−0.74	+0.8%	−1.4%

One input changed at a time; same people, same preferences. Full sample, 6,477 persons.

Where the car trips go

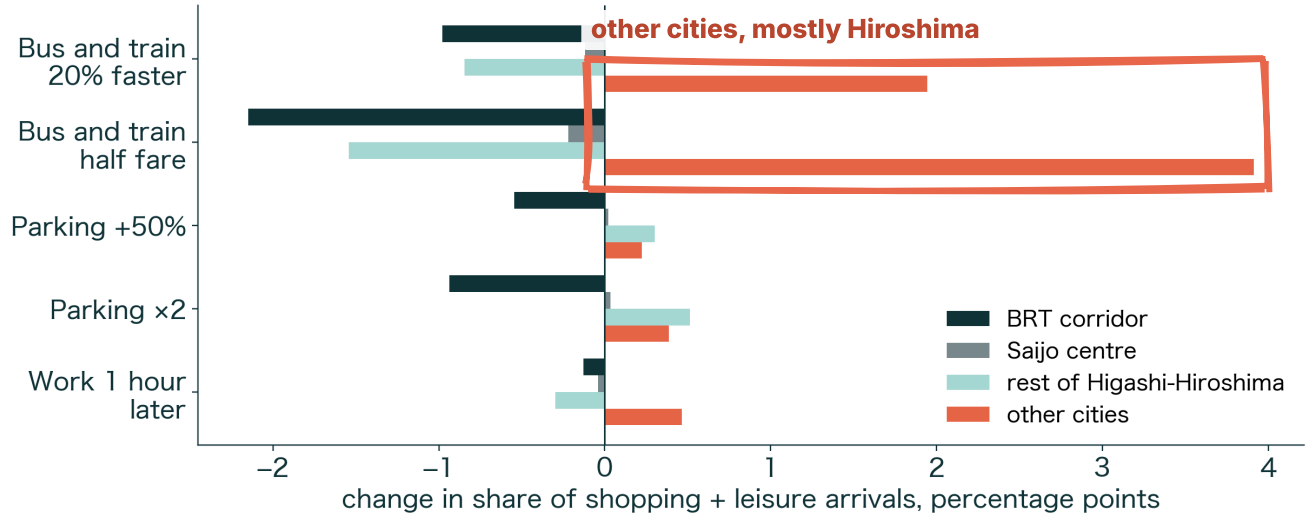
Car trips move mostly to the train, plus new train trips; dearer parking simply drops outings.



Change in trips per day, per 1,000 persons, by group: exact expected trips.

Where free-time trips go under city-wide policies

Faster or cheaper bus and train send new trips to Hiroshima by train; parking drops central visits.



Faster or cheaper bus and train: new train trips to Hiroshima (+71 a day at half fare)

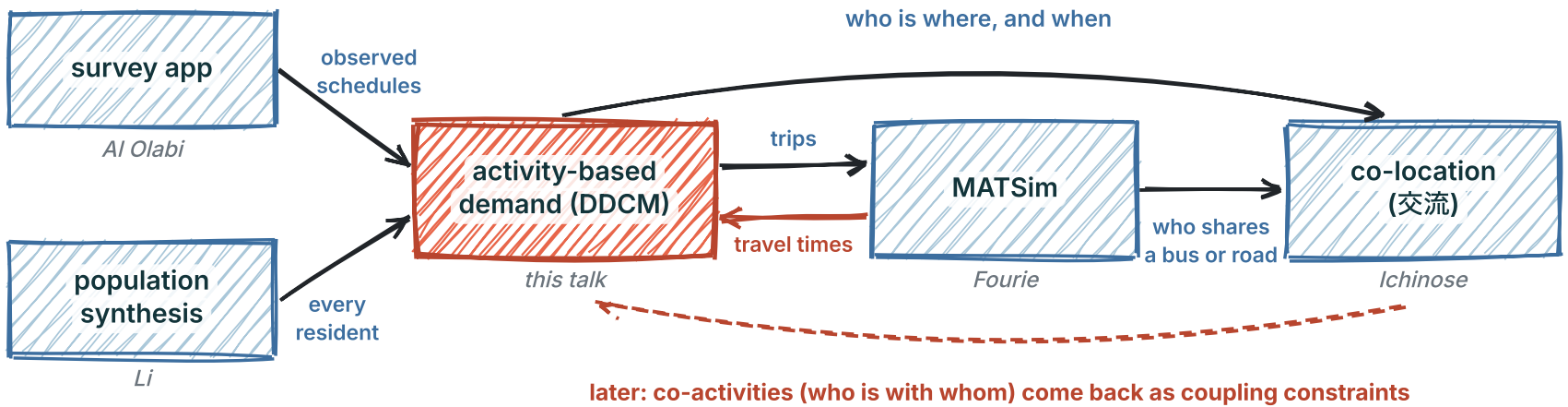
Parking ×2: centre loses ~34 visits a day; most are simply not made

PART 7

Integration and next steps

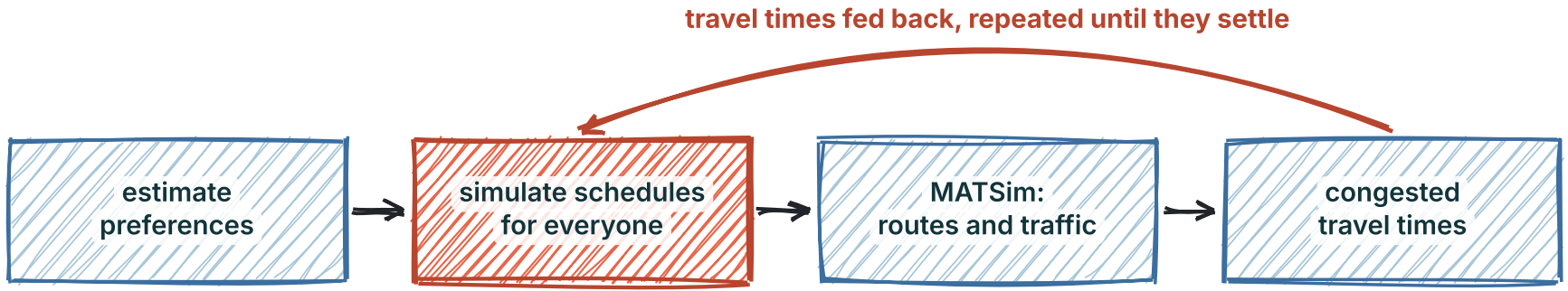
If the simulator is integrated: where this model sits

If the parts are connected: the demand model learns from the survey app, plans every resident's day, and trades trips and times with **MATSim**.



a loop in both directions: the DDCM feeds co-location, and co-location can feed the DDCM

The DDCM fills what MATSim needs

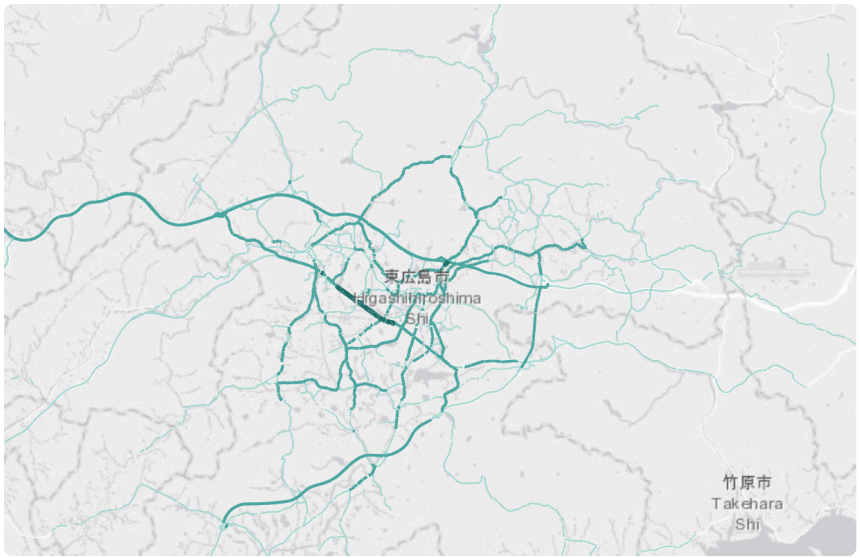


preferences never move in the loop; only the environment does

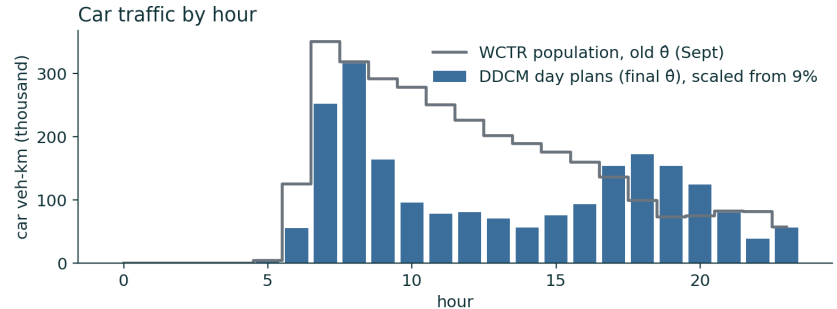
MATSim keeps the network; the DDCM keeps the choices: each does the part the other cannot

needed for: congestion, BRT with dedicated lanes, vehicle-km and emissions

First test: the model’s day plans in MATSim



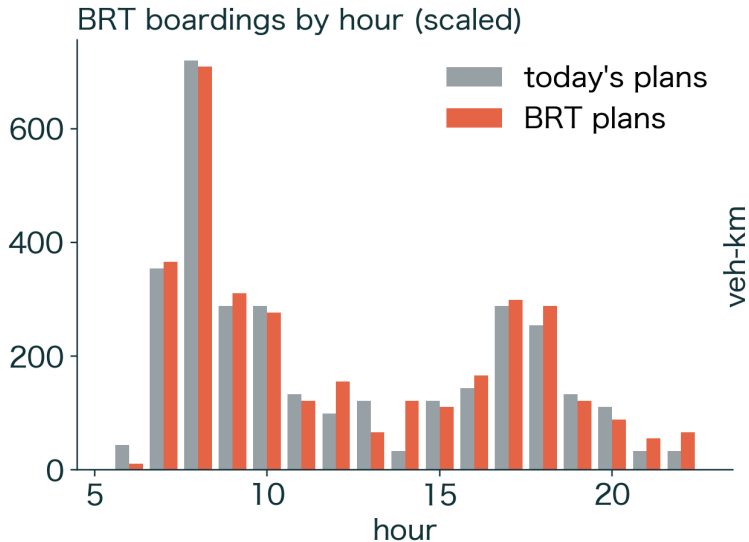
Cars per day on each road, scaled to the population: thin and light, 300 to 2,000; thick and dark, 15,000 or more.

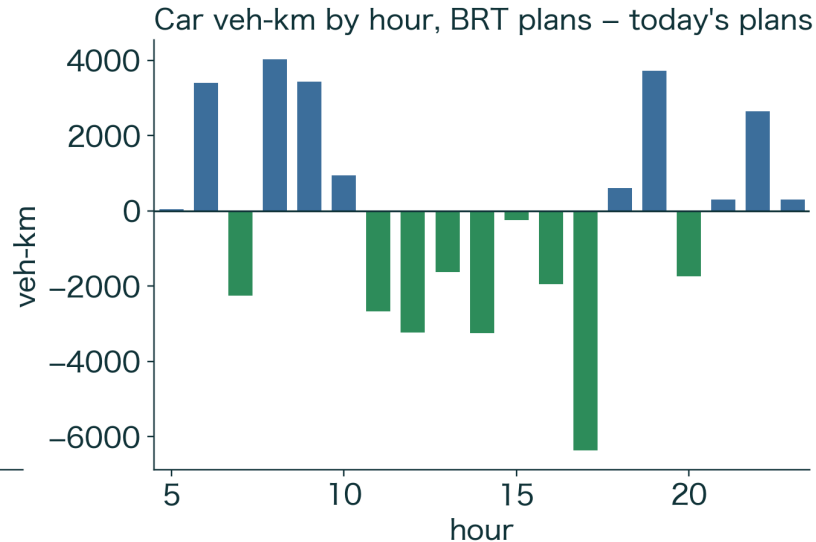


Day plans loaded	19,350 person-days (9% sample)
Car trips per day, scaled	177,530 · 28 min · 12.0 km
Link volumes vs earlier full run	correlation 0.79
Car travel times vs the diary: bias	**−0.06 min** (static −6.2)
Bus and train trips MATSim can carry	1,536 of 4,421

Routes only (mode and times are the model's); one pass, 9% sample. Not a policy result.

The BRT in MATSim: the chain runs end to end



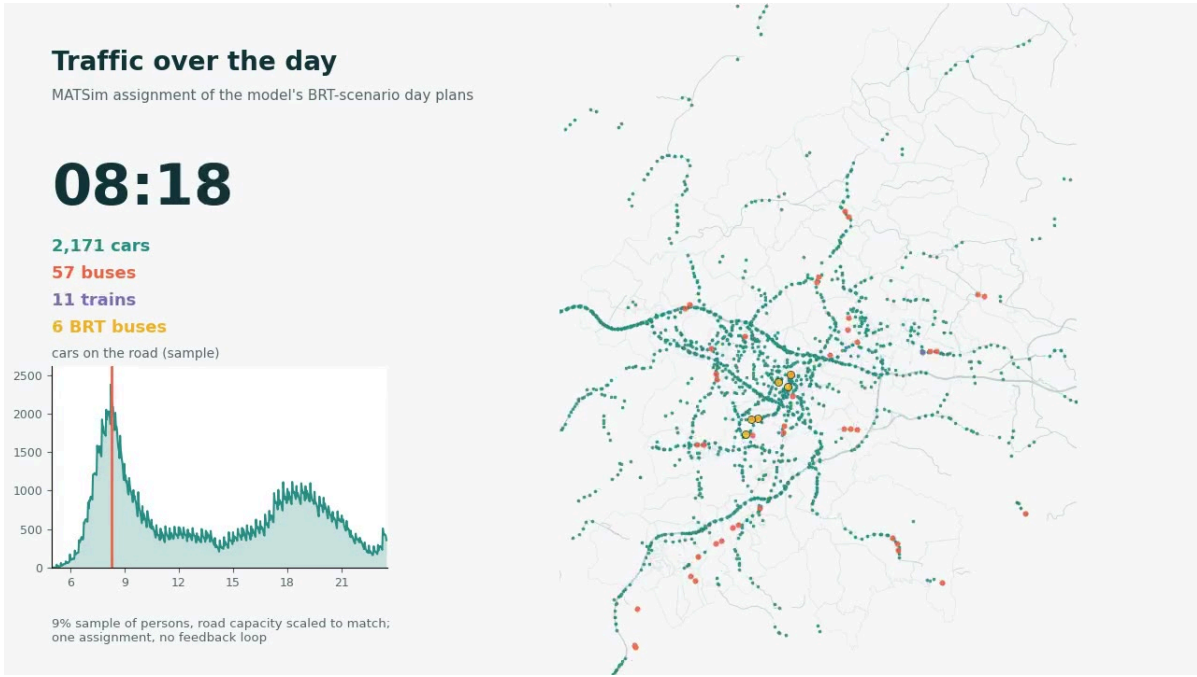


People board the BRT, mostly at Saijo station in the morning peak: a little more BRT use, a little less car traffic.

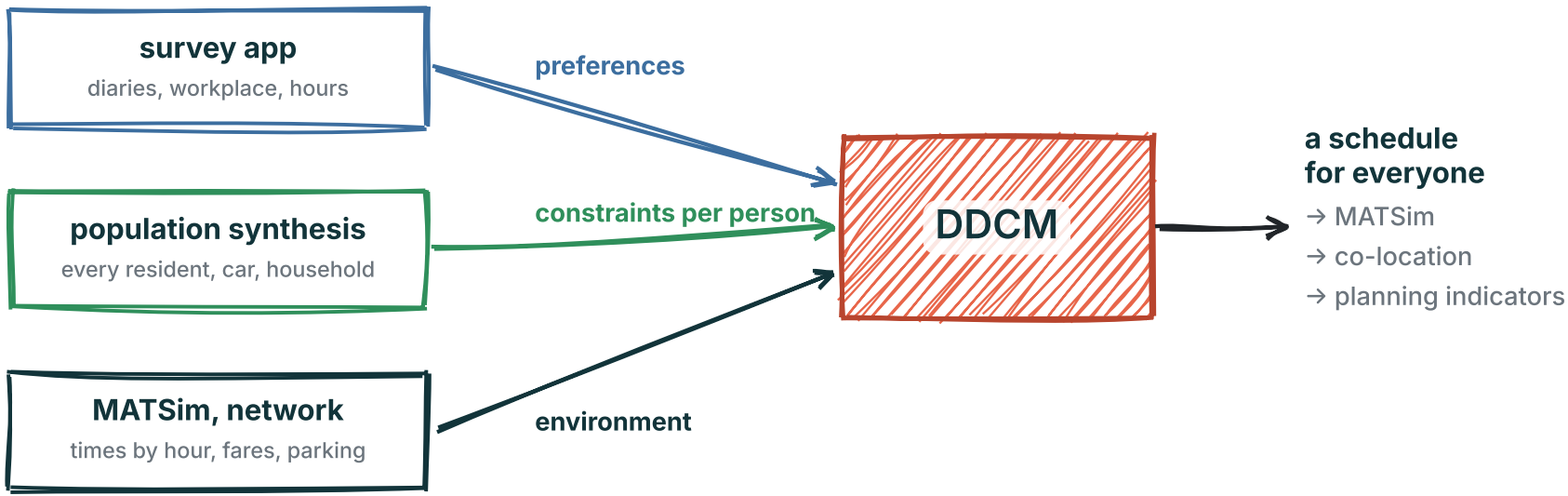
> **Not a ridership forecast:** 9% sample, one pass; MATSim carries only a third of bus and train trips.

Traffic over the day, with the BRT

▶ Animation in the live talk: final frame shown



What the demand model needs from the other parts



What it cannot do yet

The behaviour

One weekday per person
no week, no habit, no learning

Work duration from the diary
not chosen; start time as observed

Individuals, not households
shared cars and joint activities not coupled

Same preferences for everyone
differences come from constraints and attributes

The network and the BRT

Road times are static
unless the MATSim loop is run

BRT = faster buses only
no lane, congestion, comfort or reliability

BRT counted inside bus or train
BRT alone = bus; as a feeder = train (main mode)

The estimation

One iteration, not the final model
ages 13+, reference and fixed values are current choices

Shopping destinations too spread
no store size; opening hours and shop data incomplete

Bus and train time value assumed
from a trip-level model

Point estimates
no uncertainty bands on scenario effects yet

Next: a richer specification (modes, store size, what the BRT offers beyond time) and a closed loop with MATSim (lanes, congestion, the full population).

Conclusions: what each objective delivered

Objective	What we showed	Status
1 · Each person's whole day	Values of time by mode (car 79, walk 65, bicycle 48 ¥/min). The model reproduces the diaries' modes, rhythm of the day, trip counts and leisure destinations	One iteration, estimated. Shopping goes too far: 18% to other cities against 4% in the diary
2 · Constraint heterogeneity	6,451 persons, each with their own constraints, estimated together on a handful of shared maps	Done. Open: a constraint set by another person's choice
3 · Policy, person by person	BRT: about 1 ¥ per person-day on average, about 2 ¥ for people without a car. City-wide policies compared on the same people	Full sample. The BRT is a travel-time change only; next: richer specification, MATSim loop, full population
Link to MATSim	Base and BRT day plans run end to end	First test, 9% sample

1 · Trips are linked through activities, and policies act on the whole schedule.

2 · Constraint heterogeneity is managed, not ignored: every person keeps their own constraints, so who gains and who loses from a policy can be read person by person.

3 · The model is estimated from local diaries and is designed to feed MATSim and co-location.

Thank you · ありがとうございました

Words used in this talk

Travel demand model	A model of who travels, where, when and by which mode, used to test changes to the city before they are built.
Trip-based / activity-based	A trip-based model predicts each trip on its own; an activity-based model predicts the activities of the day, and the trips follow from them.
Discrete choice model	A model of a person choosing one option from a list; more attractive options are more likely, never certain.
Preference	How much people value something, such as how strongly they dislike travel time or cost. Shared, and learned from diaries.
Dynamic (DDCM)	Each choice is valued by how attractive it is now and by what it leaves possible later in the day.
Estimation / simulation	Estimation learns the preferences from observed diaries; simulation uses them to produce schedules for everyone, today or under a policy.
Constraint	A limit on what a person can do: work hours, a pick-up, opening hours, no car.
Space-time prism	Every place a person can reach between two fixed appointments, given how fast they can travel.
Shared map / mask	A shared map holds the options for many people and is calculated once; a mask is one person's list of options they cannot take, switched off on that map.
Accessibility	The value of everything a person could do in the day, given their constraints; it can also be written in yen.
MATSim	A traffic simulation that moves every trip through the road and bus network and returns travel times.

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- Bowman, J. L., Ben-Akiva, M. E. (2001). Activity-based disaggregate travel demand model system with activity schedules. *Transportation Research Part A*.
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BACKUP

Full population: already run once

Backup: the full population has been run, with an earlier estimate

September 2026 (MATSim User Meeting): every resident simulated and assigned, with an **earlier estimate**. Shows it can be done; numbers not comparable.

214,216
persons, one day each

45 min
to solve every person's day, on one CPU machine

562,731
trips assigned by MATSim on the real network

99%
of travelling persons complete their day in MATSim

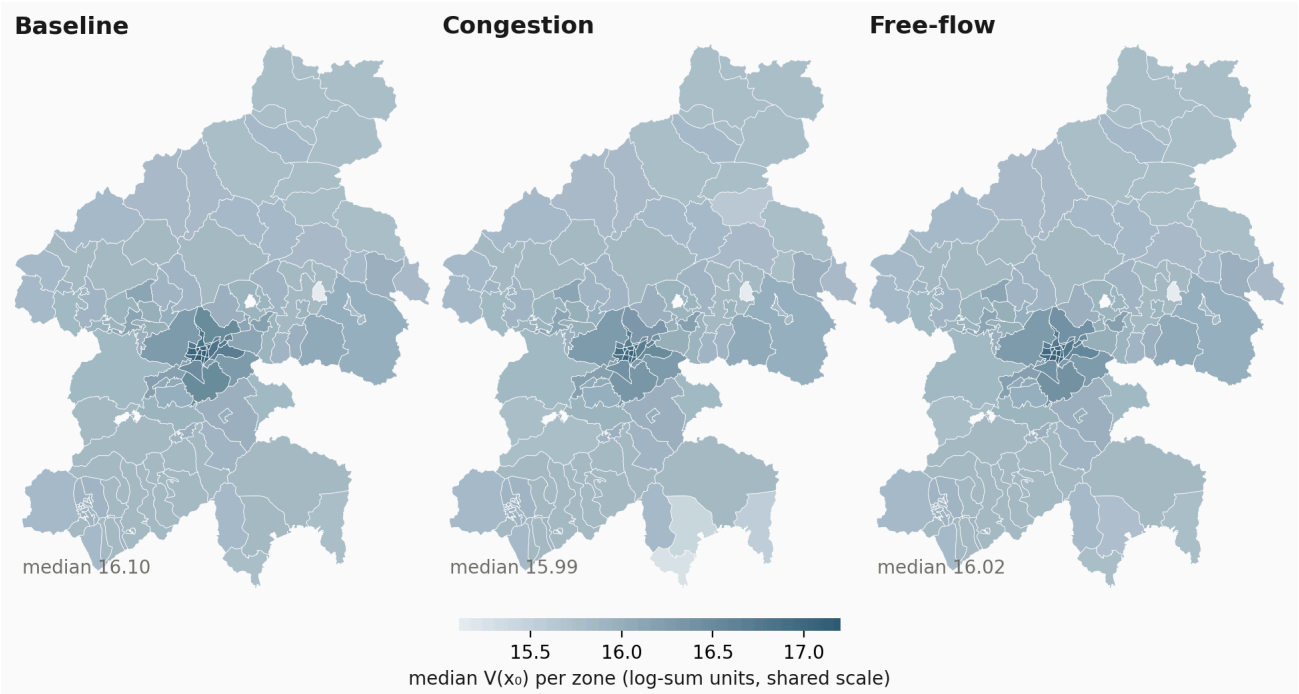
Then (September)

- Estimate of August 2026, older specification
- Older bus and train times (station to station)
- MATSim chose routes only, as now

Now (this talk)

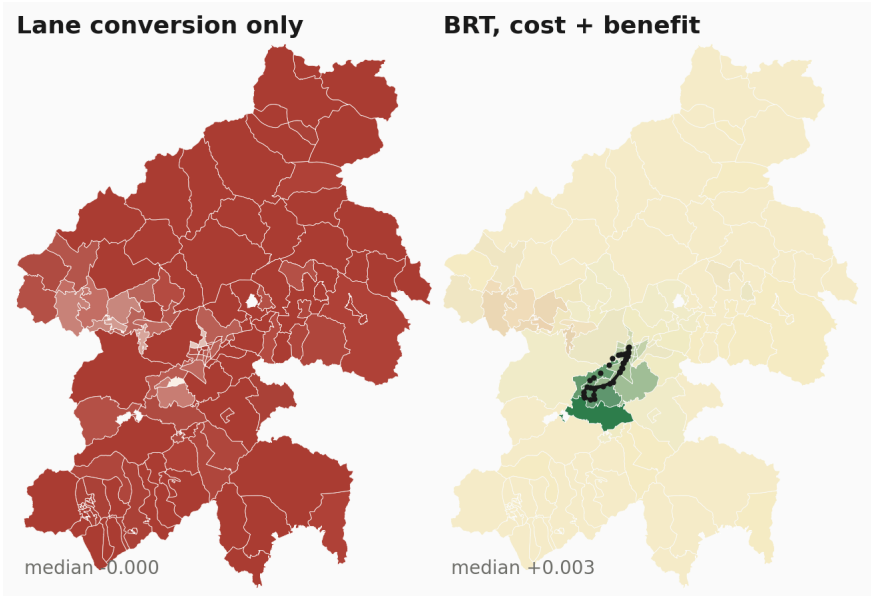
- Current estimate, robust standard errors
- Door-to-door bus and train times
- Full sample, 6,477 persons; full population is the next run

Backup: full-population accessibility, by home zone



All 214,216 residents, earlier estimate. Median accessibility by home zone, three travel-time settings.

Backup: full-population BRT, taking a car lane



Earlier estimate. **Left:** lane taken, no service: all lose a little. **Right:** with the BRT, 96.6% of corridor residents gain. Today's BRT takes no lane.